

Suboot

Federated Clinical Evidence from South Asia

Real-world patient evidence for global pharma — while no patient data ever leaves the hospital.

The Problem

Regulatory push for diversity, structural exclusion

- FDA Diversity Action Plans (FDORA) and EMA guidance now require pharma to include diverse populations in clinical trials.
- South Asia carries ~25% of the world's disease burden but supplies under 5% of trial participants.
- The data exists: Pakistani hospitals hold years of real treatment outcomes.
- Pakistan's 2025 Personal Data Protection Act makes exporting patient data illegal — the old CSV route is dead.

Why it matters: global real-world evidence (RWE) market is \$6B+ and growing ~13%/yr — and pharma is forced to look to new geographies.

The Insight

What everyone else misses

- The data-protection law didn't kill the market — it created a moat.
- The real bottleneck is trust and verifiable provenance, not data or legality.
- Pharma won't buy what it can't prove. Hospitals won't share what they can't control.
- Solve the proof problem, and the market unlocks.

Suboot's thesis: buyers pay for evidence receipts (proof), not for the data itself.

The Solution

A federated evidence platform — queries move, data never does

- Pharma uploads an analysis query, not data.
- The query runs inside each hospital's own environment.
- A custom LLM pipeline converts unstructured Urdu / Roman-Urdu / English clinical notes into standard OMOP structures, on-site.
- Only differentially-private aggregate statistics leave the hospital.
- Every query, consent, and result is written to an append-only audit ledger with zero-knowledge verification.
- The buyer pays for a tamper-evident 'evidence receipt' — proof the cohort is real, consented, and computed on genuine patients.

How It Works

1

Upload

Pharma defines the cohort and analysis (query in, no data out).

2

Structure

On-site LLM converts Urdu/English notes to OMOP locally.

3

Compute

Federated compute runs analysis inside each hospital.

4

Prove

Append-only ledger + ZK proofs mint an evidence receipt.

5

Pay

Buyer pays per verified receipt. No PHI ever moves.

This combination — federated + ledger + local-language LLM + South Asia — has never been assembled.

Product Status

Pre-prototype, architecture complete

- Concept and system architecture fully documented.
- Public research repo: github.com/zaynabtauqeer/subboot
- MVP roadmap: federated query orchestration → Urdu/English structuration → ledger verification.
- Demo target: 90 days after cohort start.

Honest stage: idea. Traction being built in the next 90 days.

Market

Pull from regulation, priced like evidence

- Global RWE market: \$6B+, ~13% CAGR (Grand View Research / Precedence, 2025).
- Every global pharma with a Diversity Action Plan is a potential buyer.
- Pricing: \$20–50 per matched de-identified record (published RWE norms).
- 5 hospitals × ~20,000 records/yr → \$2–5M potential; discounted to ~\$1.5M SOM for the first 24 months.

Competition

Who does what today — and where they fall short

- Manual workaround: CROs request CSV exports (now legally refused), hire local chart-review teams, or drop South Asian cohorts.
- TriNetX — federated networks, but no ledger, no South Asian source hospitals, no local-language extraction.
- Aetion / PicnicHealth / Flatiron — US/EU records only.
- IQVIA-style CROs — expensive, US/EU-centric services.
- Open-source federated frameworks — code, not products: no buyer, no provenance.

Unfair Advantage

Why we win

- Proprietary Urdu/English clinical-note structuration pipeline — a corpus foreign competitors can't obtain without our on-ground hospital network.
- First-mover building the South Asian hospital network; each hospital onboarded raises switching costs.
- Only vendor shipping regulator-grade (ALCOA+-style) ledger-verified evidence provenance.
- Unbuildable by a US startup (no ground network) and by a local app shop (no crypto/federated depth).

Business Model

- Buyer: global pharma and CROs with RWE and diversity mandates — they pay per evidence receipt.
- Supplier: Pakistani teaching hospitals earn revenue per cohort (supply-side unlock).
- Recurring: annual cohort contracts, not one-off sales.
- Scaling: platform, not service — marginal cost per customer approaches zero.

Traction & Roadmap

Idea stage, 90-day sprint

- Done: concept, architecture, regulatory validation, market research, public repo, full NIC application.
- Days 1–30: working federated-query demo on synthetic data.
- Days 31–60: Urdu/English note-structuration prototype; hospital conversations.
- Days 61–90: ledger-verified evidence receipt; first pilot LOI.

The Ask

Mentorship: help validating Suboot with Pakistani hospitals and global pharma/CROs, and structuring the first pilot — a mentor with RWE or health-tech deal experience.

NIC cohort access, network, and cloud credits to ship the demo.

github.com/zaynabtauqeer/suboot